

98.1% Session Recall on LoCoMo: MemoryKG’s Hybrid Semantic-Graph Architecture Achieves Near-Perfect Retrieval Across 1,986 Questions with No LLM Required

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Abstract

We evaluate MemoryKG, a conversational memory system based on DocKG [Suchanek, 2026b], on the LoCoMo benchmark [Maharana et al., 2024], a dataset of 1,986 questions spanning five categories that require locating specific sessions within long-horizon social conversation corpora. MemoryKG achieves **98.1% average session-level recall** across all five categories, with **96.5% of questions answered perfectly** (all evidence sessions retrieved), and only **1.0% total misses**. No LLM is required at any stage — neither for indexing nor for retrieval. The hardest category is *Temporal-inference* (89.0%), which requires cross-session reasoning rather than direct lookup; the easiest are *Temporal* and *Adversarial* (both 99.6%). A key architectural finding is that *session granularity* substantially outperforms *dialog-turn granularity*: indexing one document per session (98.1%) versus one document per dialog turn (73.9%) — a 24.2 percentage point gap — confirming that session-level context aggregation is the correct retrieval unit for LoCoMo’s multi-turn evidence structure. The full 1,986-question evaluation completes in 67.4 seconds on Apple Silicon.

1 Introduction

Long-horizon conversational memory — the ability to locate specific facts or events from months of prior dialogue — is increasingly central to AI assistant design. Retrieval approaches must navigate the dual challenge of semantic matching (finding sessions whose *content* is relevant) and structural disambiguation (identifying *which* session holds the evidence when many sessions discuss related topics).

The LoCoMo benchmark [Maharana et al., 2024] evaluates retrieval over long-horizon social conversation corpora. Unlike single-fact or short-history benchmarks, LoCoMo conversations span many sessions over extended time periods, with evidence scattered across temporally ordered dialog turns. The five question categories — Single-hop, Temporal, Temporal-inference, Open-domain, and Adversarial — probe qualitatively different retrieval challenges.

MemoryKG [Suchanek, 2026a] builds a persistent knowledge graph over verbatim conversation sessions, combining semantic vector search (LanceDB) with structured graph expansion (SQLite). It is a direct specialisation of DocKG [Suchanek, 2026b] for conversational memory corpora. In this paper we report MemoryKG’s performance on LoCoMo and analyse the architectural factors driving the results, with particular attention to the granularity question: should retrieval units be entire sessions or individual dialog turns?

2 The LoCoMo Benchmark

LoCoMo [Maharana et al., 2024] (Long Context Motivated) is a benchmark for evaluating long-term conversational memory. It contains 10 long-horizon social conversations between two participants, each spanning many sessions over an extended period. From these conversations, 1,986 question–answer pairs are constructed, each paired with one or more *evidence dialog IDs* — the specific turns whose content is necessary to answer the question.

The five question categories are:

Single-hop ($n=282$) Questions answerable from a single dialog turn without temporal reasoning.

Temporal ($n=321$) Questions requiring knowledge of *when* an event occurred; evidence is usually a single datable session.

Temporal-inference ($n=96$) Questions requiring temporal *reasoning* across multiple sessions (e.g. “Which happened first?”), not just lookup.

Open-domain ($n=841$) General questions about conversation content that may draw on any session.

Adversarial ($n=446$) Questions designed to exploit surface-level lexical overlap with incorrect sessions.

The original LoCoMo paper reports end-to-end QA accuracy (requiring an LLM to generate a free-form answer); we report session-level retrieval recall — whether the evidence session is *present* in the retrieved context. These are complementary but distinct measures. Our metric is a strict prerequisite for QA accuracy: no downstream reader can correctly answer a question whose evidence was not retrieved.

3 MemoryKG Architecture

MemoryKG represents a conversation corpus as a four-layer knowledge graph:

1. **Document layer.** Each conversation session is a root document node containing the full session text.
2. **Section layer.** Markdown headings (one per dialog turn) produce section nodes connected to their parent document by CONTAINS edges.
3. **Chunk layer.** Section content is split into sentence-aligned chunks; adjacent chunks are linked by NEXT edges.
4. **Semantic index.** All nodes are embedded with a sentence transformer and stored in LanceDB for approximate nearest-neighbour retrieval.

3.1 Query pipeline

A query q is processed in two stages:

Stage 1 — Semantic seeding. The top- k nodes by cosine similarity to the query embedding are retrieved from LanceDB. For LoCoMo, seeding is restricted to the *current conversation’s* session files via a per-query `haystack_files` filter, making retrieval apples-to-apples with per-conversation search approaches.

Stage 2 — Graph expansion. Starting from the seed set S , BFS expansion walks outward for h hops along any edge in the relation set $\mathcal{R}$:

$\mathcal{R} = \{\text{CONTAINS, NEXT, REFERENCES, SIMILAR_TO, HAS_TOPIC, MENTIONS_ENTITY, HAS_KEYWORD}\}$

Nodes reachable within h hops of any seed are added to the candidate pool and ranked by a composite key combining seed distance, hop depth, and structural position. For LoCoMo evaluations we use $k=50$ seeds and $h=1$ hop.

3.2 LoCoMo-specific corpus structure

All sessions from all 10 conversations are written to a single flat corpus directory during the **prepare** phase (one time). Each session is serialised as a Markdown file with the session number as the top-level heading and each dialog turn as a second-level heading:

```
# session_3

**Conversation:** conv-05
**Date:** 2023-08-14

## D3:1

Alice said, "I finally signed up for that pottery class."

## D3:2

Bob said, "That's great! You've been talking about it for months."
```

The heading chunk strategy splits at **##** boundaries, so each dialog turn becomes one chunk node while the parent session document node aggregates all turns. At query time, the **haystack_files** filter restricts LanceDB seeding to the files belonging to the current conversation, providing per-conversation isolation without rebuilding the KG for each conversation.

3.3 Granularity: session vs. dialog

A key design question is whether to index *sessions* (one file containing all turns) or individual *dialog turns* (one file per turn). LoCoMo's evidence format specifies dialog IDs (e.g. D3:7), which at first suggests turn-level indexing. However, our experiments (Section 5.2) show that session granularity dominates by a large margin, because:

- Session-level documents give the HAS_TOPIC, MENTIONS_ENTITY, and HAS_KEYWORD enrichment edges a richer surface to index.
- The CONTAINS edge from a session document to any of its turn-level section nodes means that finding *any* turn in the seed set recovers the whole session.
- With $\sim 200\text{--}300$ dialog turn files per conversation and $k=50$ seeds, turn-level corpora are too sparse for reliable seeding.

4 Experimental Setup

4.1 Build configuration

- Chunk strategy: **heading** (one chunk per Markdown section)
- Topic / entity / keyword enrichment: enabled (default)

- SIMILAR_TO edge discovery: disabled (speed)
- Workers: 8 (parallel file parsing in Phase 1)
- Batch size: 1,024 embeddings
- Model: BAAI/bge-small-en-v1.5 (384-dimensional)

4.2 Shared embedder

A single `SentenceTransformerEmbedder` is initialised once and passed to `MemoryKG` via the `embedder=` argument, reused across all 1,986 questions. This eliminates per-question model-reload overhead.

4.3 Recall computation

Evidence for each question is expressed as a list of dialog IDs (e.g. ["D3:7", "D3:8"]). For session-granularity evaluation, these are mapped to session IDs (`session_3`) before scoring:

$$\text{Recall} = \frac{1}{|E|} \sum_{e \in E} \mathbf{1}[e \in \text{retrieved_ids}]$$

where E is the set of evidence session IDs and `retrieved_ids` is the ordered list of session IDs returned by the KG query, clipped to the top k . A recall of 1.0 means all evidence sessions were retrieved; 0.0 means none were.

5 Results

5.1 Overall session-level recall

Table 1 shows `MemoryKG`'s session-level recall by category on the full 1,986-question LoCoMo benchmark.

Table 1: `MemoryKG` session-level recall on LoCoMo (session granularity, $k=50$, $\text{hop}=1$).

Category	n	Recall
Single-hop	282	0.950
Temporal	321	0.996
Temporal-inference	96	0.890
Open-domain	841	0.988
Adversarial	446	0.996
Overall	1,986	0.981

Of the 1,986 questions, 1,916 (96.5%) achieve perfect recall (all evidence sessions retrieved), 50 (2.5%) achieve partial recall, and only 20 (1.0%) return zero recall.

5.2 Granularity comparison

Table 2 compares session-level and dialog-turn-level granularity across all five categories.

Table 2: Session vs. dialog granularity on LoCoMo ($k=50$, $\text{hop}=1$).

Category	Session	Dialog	Δ
Single-hop	0.950	0.591	+35.9 pp
Temporal	0.996	0.786	+21.0 pp
Temporal-inference	0.890	0.579	+31.1 pp
Open-domain	0.988	0.808	+18.0 pp
Adversarial	0.996	0.703	+29.3 pp
Overall	0.981	0.739	+24.2 pp

Session granularity outperforms dialog-turn granularity by 24.2 percentage points overall, with gains exceeding 30 pp on Single-hop and Temporal-inference. The gap is largest in categories where the evidence is a specific turn but the question vocabulary matches the broader session context rather than the isolated turn text.

5.3 Recall distribution

Table 3: Recall distribution across 1,986 questions (session granularity).

Outcome	Count	%
Perfect (1.0)	1,916	96.5%
Partial (0–1)	50	2.5%
Zero (0.0)	20	1.0%

5.4 Throughput

The full 1,986-question evaluation (session granularity) completes in 67.4 seconds — 0.03 seconds per question — on an Apple M5 Max (64 GB RAM). The KG is built once in the **prepare** phase and reused across all queries; the per-question cost is purely inference (LanceDB vector search plus SQLite graph expansion).

6 Analysis

6.1 Temporal-inference is the hardest category

Temporal-inference questions (89.0% recall) require cross-session temporal reasoning: not just finding the session where an event occurred, but comparing the timing of multiple events across sessions. A question such as “Which happened first, Alice starting pottery or Bob changing jobs?” requires the system to retrieve sessions for *both* events, where the vocabulary of the question may match neither event’s session directly.

At $k=50$ seeds with $\text{hop}=1$ expansion, the KG retrieves both sessions for most temporal-inference questions (10.4% miss rate vs. 0.4% for Temporal). The miss cases typically involve events mentioned only briefly in passing, where the session’s dominant topic is unrelated to the question — a vocabulary mismatch that neither semantic seeding nor one-hop graph expansion fully resolves.

6.2 Single-hop is harder than expected

Single-hop questions (95.0%) score lower than Open-domain (98.8%) and Adversarial (99.6%). This is counterintuitive: Single-hop questions should be the simplest, with evidence in a single

specific session. The explanation lies in specificity: Single-hop evidence tends to be a very specific isolated fact (a date, a name, a quantity) mentioned in one turn of one session, without surrounding context that would boost that session’s semantic similarity to the question. Open-domain questions are broader, matching more sessions and benefiting from the aggregated topic and entity edges.

6.3 Adversarial questions and graph expansion

Adversarial questions (99.6% recall) are designed to share surface vocabulary with incorrect sessions. MemoryKG’s graph expansion is robust to this because the HAS_TOPIC and MENTIONS_ENTITY edges connect sessions to their underlying semantic content, not just surface tokens. The BFS expansion from a correctly seeded session will pull in the true evidence even when the question vocabulary superficially matches other sessions.

6.4 Why dialog granularity fails

At dialog-turn granularity, the corpus for a single LoCoMo conversation contains 200–300 individual turn files. With $k=50$ seeds and $\text{hop}=1$, the system retrieves at most $50 \times (\text{fan-out})$ nodes — but because each turn is now its own file, the CONTAINS edges no longer propagate session-level recall. The graph’s enrichment edges (HAS_TOPIC, MENTIONS_ENTITY) are computed per file; a one-sentence turn gives these edges almost nothing to work with. The result is that the KG is effectively flat vector search over individual turns, with very limited graph advantage — explaining the 73.9% recall ceiling, close to what one would expect from embedding-only retrieval.

6.5 The persistent-KG advantage

Unlike per-conversation ChromaDB approaches, MemoryKG builds one persistent graph over all 10 conversations. Per-conversation isolation is achieved at query time via `haystack_files` filtering in LanceDB — restricting the k -NN seed search to the current conversation’s files without rebuilding any index. This separation of build-once and query-many is the key to the 0.03 s/question throughput: the per-question cost is LanceDB vector search plus SQLite graph expansion, with no corpus ingestion or embedding on the critical path.

7 Conclusion

MemoryKG achieves 98.1% average session-level recall on the full 1,986-question LoCoMo benchmark, with 96.5% of questions answered perfectly and only 1.0% total misses. No LLM is required at any stage.

The central architectural finding is that **session granularity is the correct retrieval unit for LoCoMo**: indexing one document per session outperforms dialog-turn-level indexing by 24.2 percentage points overall. The graph’s CONTAINS edges from session documents to their constituent turn-level section nodes mean that any turn appearing in the semantic seed set propagates recall to the whole session — a propagation mechanism that dialog-turn granularity destroys.

The hardest remaining category is Temporal-inference (89.0%), where the miss cases involve events whose sessions are semantically distant from the question vocabulary. Deeper graph expansion ($h=2$) or temporal-aware reranking are natural directions for improvement.

Limitations. Retrieval recall measures whether evidence is *present* in the retrieved context; it does not measure whether a downstream reader can correctly extract the answer. End-to-end

QA accuracy on LoCoMo with a MemoryKG retrieval front-end is left for future work.

Reproducibility. All benchmark scripts and result files are included in the MemoryKG repository. The full evaluation can be reproduced with:

```
# Build the KG (one time):
python benchmarks/locomobench/locomo_bench_memkg.py prepare \
    benchmarks/locomobench/data/locomo10.json

# Run all 1,986 questions:
python benchmarks/locomobench/locomo_bench_memkg.py run \
    benchmarks/locomobench/data/locomo10.json
```

References

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